

# HARIS WAHEED BHATTI

BIM Specialist | Information Manager (ISO 19650)

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Open to relocating within the EU; available immediately. Currently authorized to work in Slovenia under a student permit (until 30 Sep 2026), transitioning to standard EU work authorization thereafter.

## PROFILE

Civil engineer with a European Master's in Building Information Modelling, combining ISO 19650 coordination practice with the ability to build the tooling around it. Delivered a federated multidiscipline model with clash detection and EIR compliance, and published an open-source IFC-to-Neo4j converter on PyPI.

## PROJECTS

**Federated BIM Model – Pakistani-Themed Expo, Guimarães** | IFC, Autodesk Construction Cloud ..... Oct 2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

**ifctoneo4j – Python package (PyPI)** | Python, IfcOpenShell, Neo4j, Cypher ..... 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

**BIMGraphExplorer – LLM agent for IFC question answering** | Python, MCP, Neo4j ..... 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.
- Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge; 23 IFC datasets, 367 questions, 15 projects). Overall accuracy 72.0% against the baseline's 82.3%, but reached source-transparency parity (99.7% versus 100%) and outperformed it on 5 of 23 datasets, by up to 24 points on one.

## EDUCATION

**MSc, European Master in BIM (BIM A+)** | UL FGG, Ljubljana, Slovenia ..... Oct 2025 – Jul 2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. [hdl.handle.net/20.500.12556/RUL-185169](https://hdl.handle.net/20.500.12556/RUL-185169)
- Applied coursework spanning the BIM delivery chain: IFC schema auditing and model remediation, parametric façade optimisation in Dynamo, process modelling and workflow improvement, 5D quantity take-off and cost estimation, scan-to-BIM from point-cloud capture, and energy management and analysis.

**BE Civil Engineering** | NUST, Islamabad, Pakistan. Final grade 3.25/4.00 ..... Sep 2020 – Jun 2024

- Coursework in hydrology and water resources engineering (AutoCAD, HEC-HMS, HEC-RAS), geographic information systems (ArcGIS), and urban planning.

## EXPERIENCE

**Contract Management Intern** | Engineering and Management Services, Pakistan ..... June 2023

- Reviewed contract reports and arbitration documentation under FIDIC conditions; presented findings to the internal legal team.

**Civil Engineering Intern** | Fatima Fertilizer Company, Pakistan ..... July 2023

- Modelled a 6,804 m<sup>2</sup> rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

**Research Intern (remote)** | Southern Illinois University Edwardsville, USA ..... July 2023

- Synthesised academic literature to support a faculty-mentored research proposal.

## SKILLS

**BIM & AEC:** Revit, AutoCAD, ArcGIS, clash detection, federated models, IFC (IFC2X3, IFC4, IFC4X3), ISO 19650, Employer's Information Requirements, Dynamo, Autodesk Construction Cloud, IfcOpenShell

**Programming & Data:** Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git, MS Office (Word, Excel)

**AI / LLM:** Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking

**Languages:** English (C1), Slovene (A1, actively learning)